CSE311L: DATABASE SYSTEMS LAB

ERD & Schema Design

Auvryon - Intelligent Digital Finance & Payment Ecosystem

Group	1
Course / section	CSE311L.3 (Section 3)
Faculty	Tns1
Submission deadline	5 November 2026, 11:59 PM
Notation	Chen (entity, relationship, attribute)

Group members

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1. Scope and consistency with the project proposal

System (from the proposal): Auvryon is a responsive PHP/MySQL web application that lets users send, request, split, escrow and track simulated money through a double-entry ledger with built-in protection against wrong transfers. All money in the project is simulated.

This design describes the database **auvryon_db** exactly as implemented: every entity, relationship and foreign key below was read from the live schema, so the diagrams, tables and logical schema cannot disagree with the project.

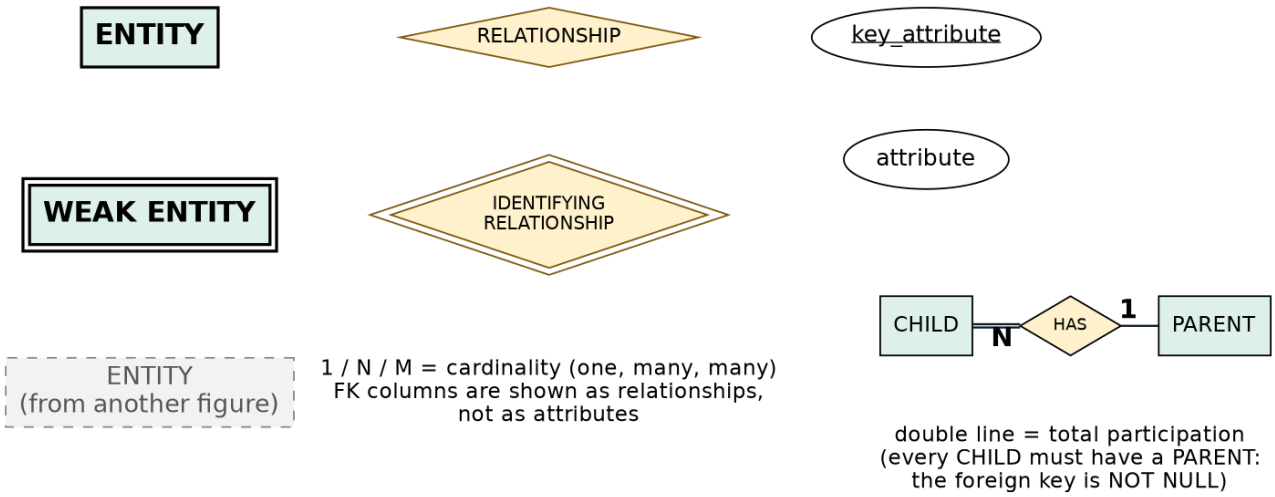
How the six core features of the proposal map to the design

Proposal core feature	Main entities (figure)
1. Digital wallet and SafePay (registration, login, wallets, transfers, requests, PINs, duplicate prevention, wrong-transfer cancellation window)	USERS, USER_SETTINGS, DEVICES, WALLETS, TRANSACTIONS, TRANSFER_HOLDS, PAYMENT_REQUESTS, IDEMPOTENCY_KEYS, RISK_ALERTS (Figures 1, 2, 3, 4)
2. Double-entry ledger and unified activity (balanced entries, searchable history, receipts, statements, balance explainer)	LEDGER_ACCOUNTS, LEDGER_ENTRIES, TRANSACTIONS, WALLETS, AUDIT_LOGS (Figures 2, 3)
3. Group finance and IOU tracking (equal / custom / percentage splits, settlements, shared budgets, IOUs with partial repayments)	EXPENSE_GROUPS, GROUP_EXPENSES, GROUP_SETTLEMENTS, BUDGETS, IOUS, IOU_REPAYMENTS and the M:N relationships MEMBER_OF and SPLIT_AMONG (Figures 5, 6)
4. Safe-to-spend and recurring payments (forecast, bill/subscription calendars, money rules, virtual tokens)	RECURRING_ITEMS, REMINDER_LOG, MONEY_RULES, RULE_RUNS, POCKETS, VIRTUAL_TOKENS, TOKEN_CHARGES (Figures 6, 7, 8)
5. Escrow, invoices and disputes (milestone escrow in a separate ledger account, invoices with payment links, QR checkout, refunds)	ESCROWS, ESCROW_MILESTONES, DISPUTES, DISPUTE_EVIDENCE, INVOICES, INVOICE_ITEMS, INVOICE_PAYMENTS, MERCHANTS, CHECKOUT_SESSIONS (Figures 4, 10, 11)
6. Pockets, guardian controls, simulated multi-currency and offline QR	POCKETS, DEPENDENTS, DEPENDENT_RESTRICTIONS, APPROVAL_POLICIES, APPROVAL_REQUESTS, EXCHANGE_RATES, OFFLINE_TOKENS, OFFLINE_QUEUE (Figures 4, 7, 8, 9)

Design in numbers

Item	Count
Relations (tables) in the logical schema	51
Entities in the ERD (strong + weak)	47 (38 strong, 9 weak)
Many-to-many relationships (each becomes a link relation)	4: MEMBER_OF, SPLIT_AMONG, APPROVER_OF, VOTES_ON
One-to-many relationships	65
One-to-one relationships	13
Total relationships	82
Foreign-key columns	86
Check	47 entities + 4 link relations = 51 relations

2. Reading the diagrams (Chen notation)



- **Entity names** are the table names in capitals (USERS = table users). **Underlined** attributes are primary keys.
- Each figure shows the primary key and up to three more descriptive attributes of every entity so the diagram stays readable. **The complete attribute list of every entity is in Section 3.** Foreign-key columns are not drawn as attributes: they are the relationships.
- **Cardinality** labels sit next to the entity they describe: the parent side of a one-to-many relationship is 1 and the child side is N. Many-to-many relationships are labelled M and N.
- **Weak entities** (double rectangle) cannot exist without their owner and are joined to it by an identifying relationship (double diamond). They keep a surrogate id as primary key plus a foreign key to the owner.
- Entities drawn with a dashed grey box belong to another figure and are repeated only to show the relationship. Figure titles are numbered 01 to 11.

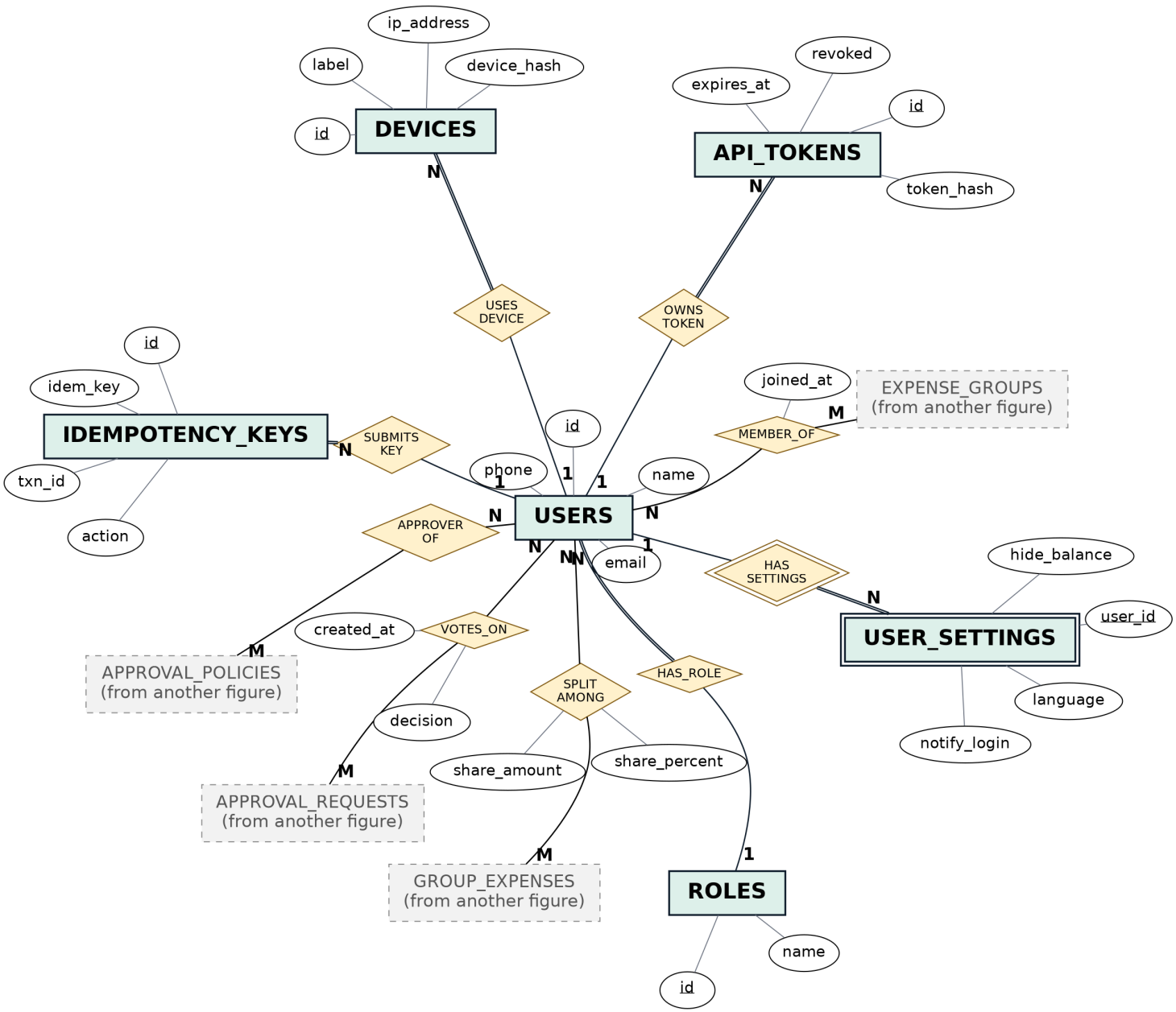


Figure 1. Entities: ROLES, USERS, USER_SETTINGS, DEVICES, API_TOKENS, IDEMPOTENCY_KEYS. Relationships shown: 9.

02. Notifications, risk alerts, audit, timelines and platform tables

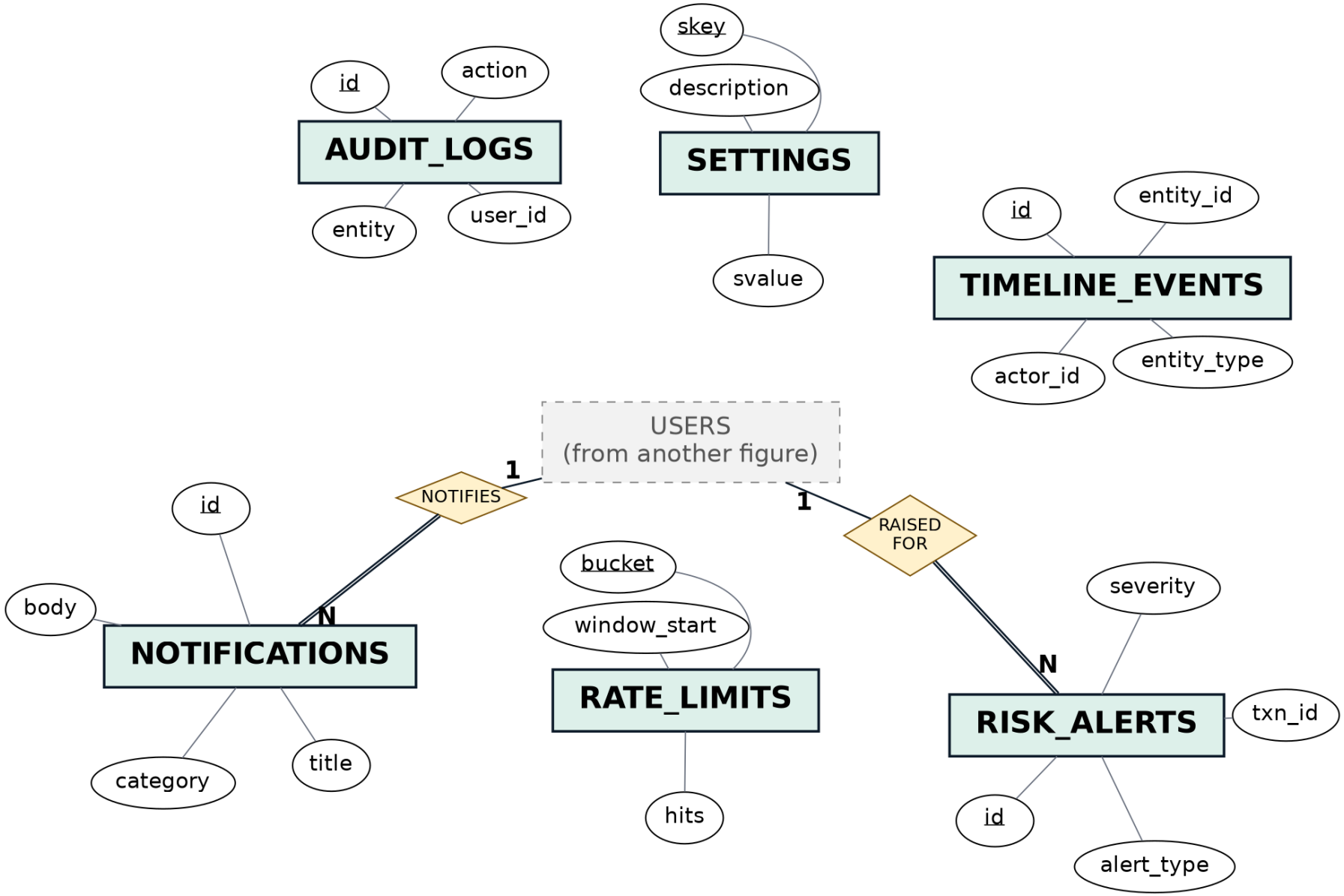


Figure 2. Entities: NOTIFICATIONS, RISK_ALERTS, AUDIT_LOGS, TIMELINE_EVENTS, RATE_LIMITS, SETTINGS. Relationships shown: 2.

03. Ledger core: accounts, wallets, transactions, entries, holds

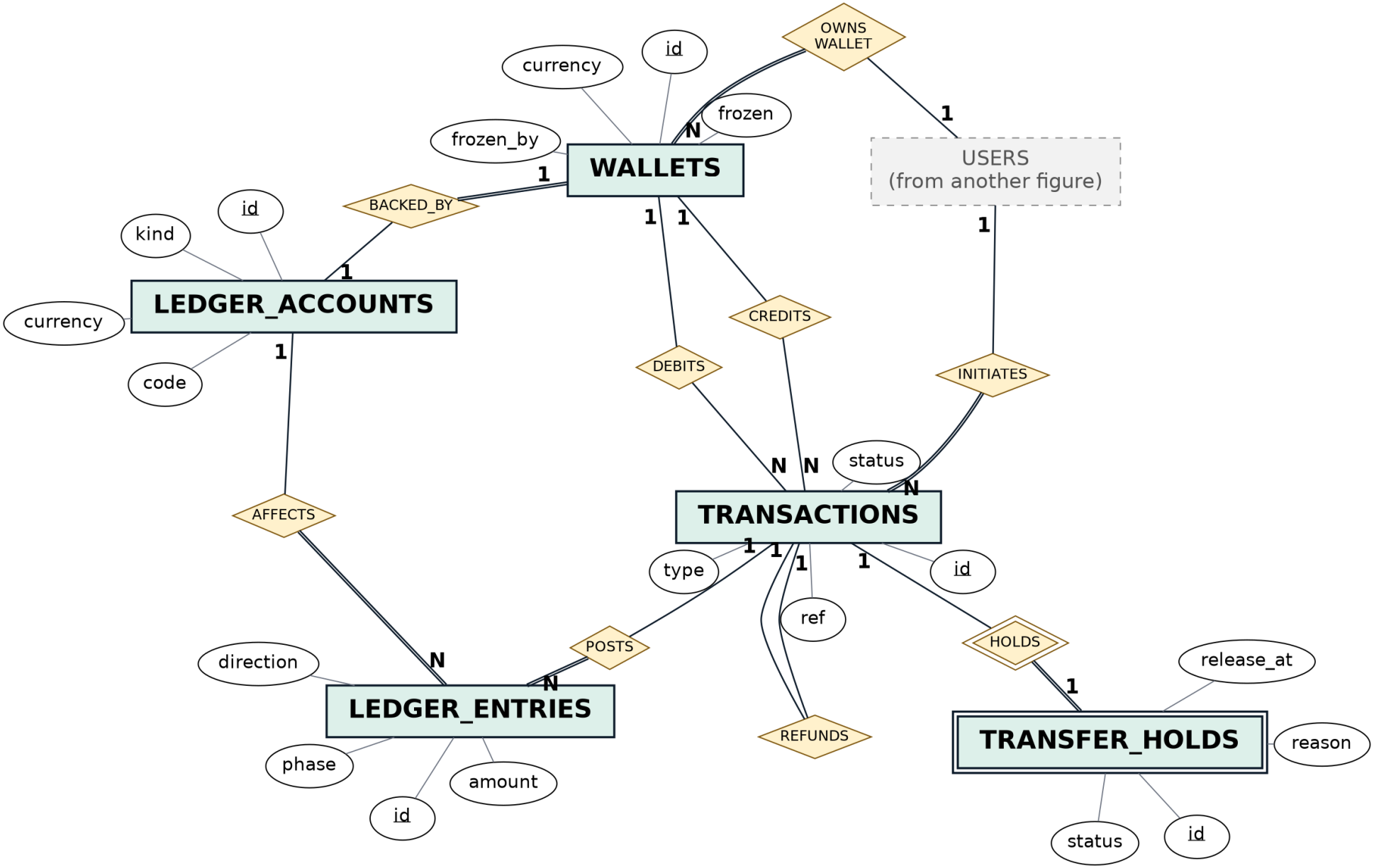


Figure 3. Entities: LEDGER_ACCOUNTS, WALLETS, TRANSACTIONS, LEDGER_ENTRIES, TRANSFER_HOLDS. Relationships shown: 9.

04. Payment requests, exchange rates, merchants, QR checkout

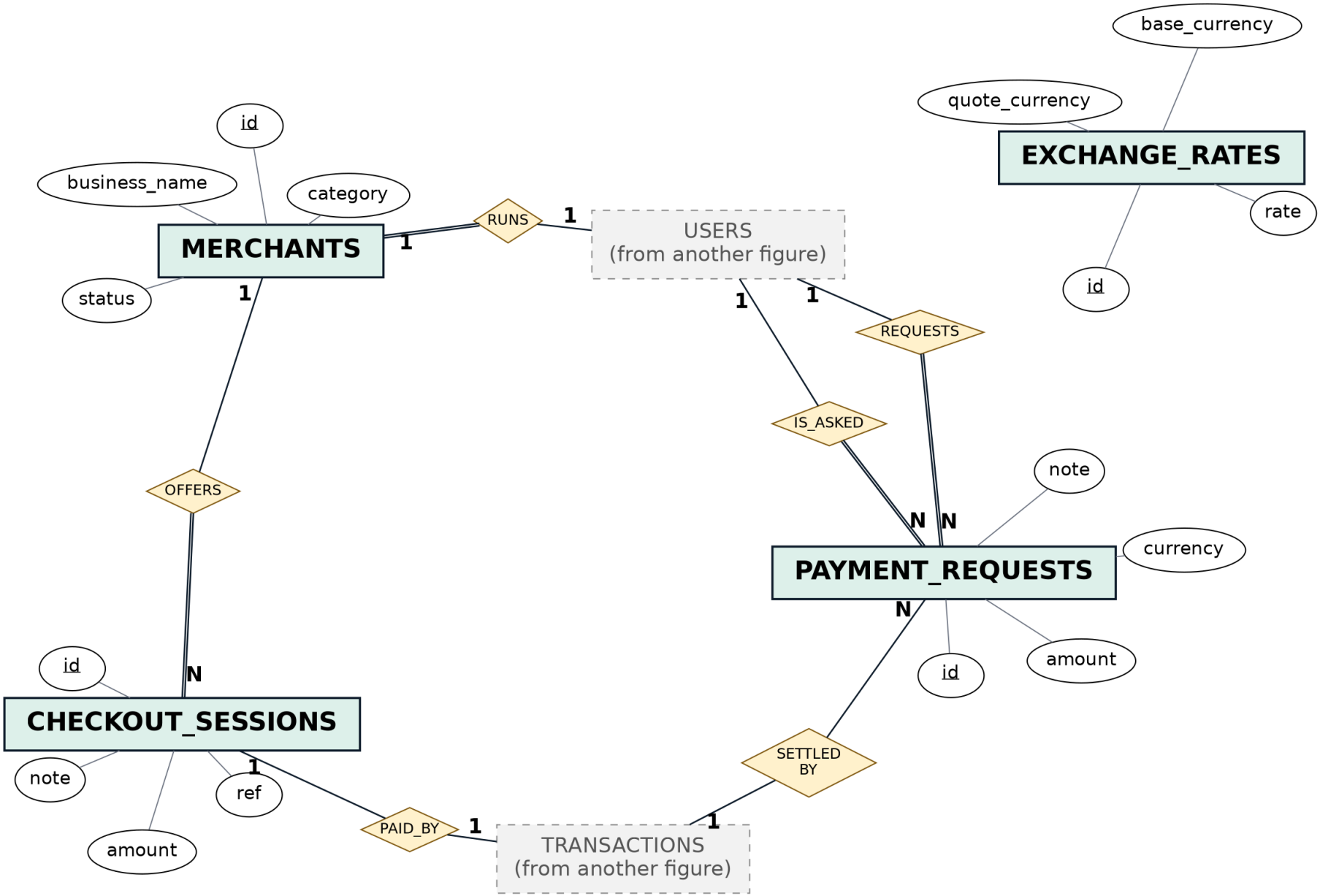


Figure 4. Entities: PAYMENT_REQUESTS, EXCHANGE_RATES, MERCHANTS, CHECKOUT_SESSIONS. Relationships shown: 6.

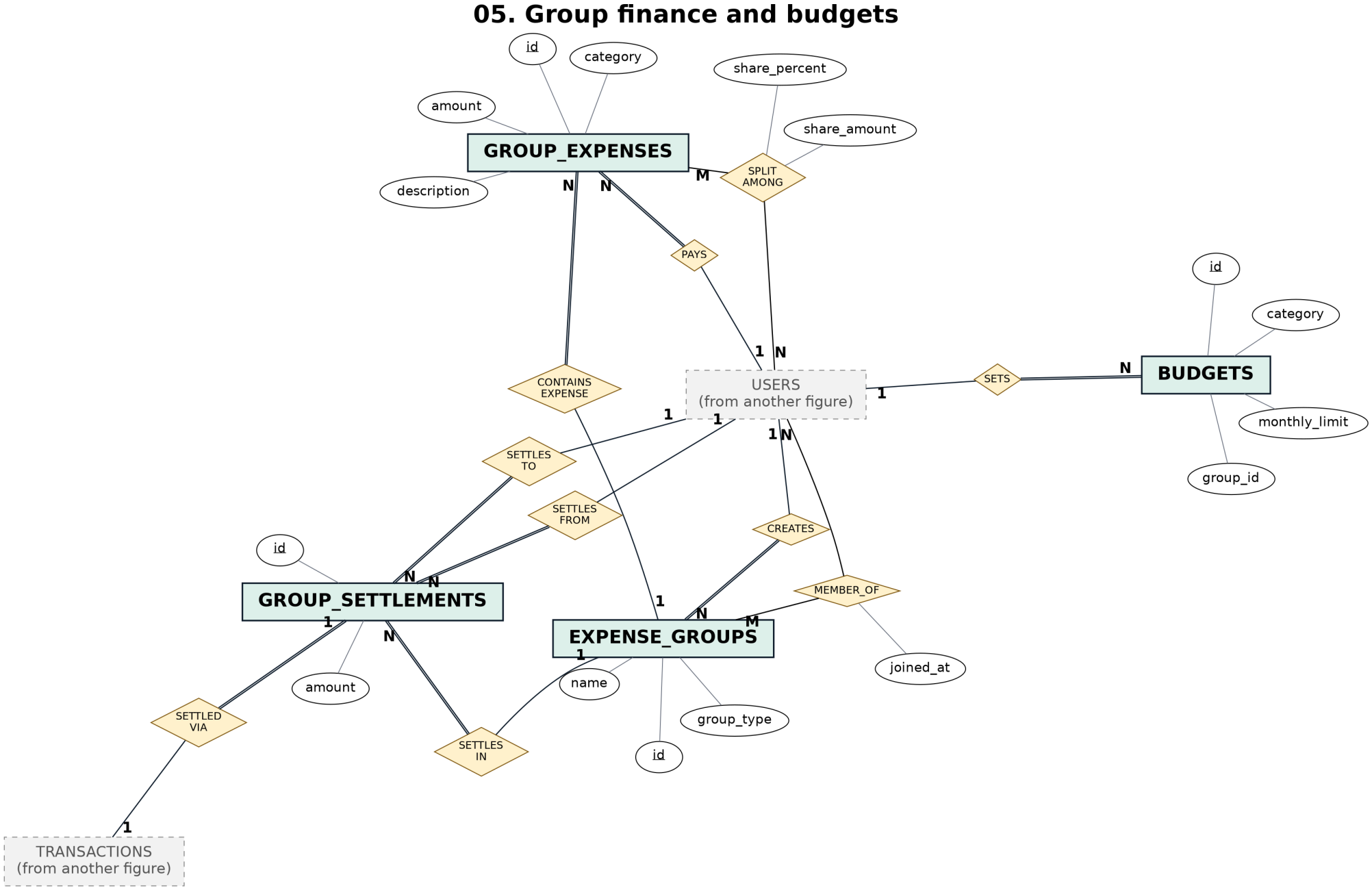


Figure 5. Entities: EXPENSE_GROUPS, GROUP_EXPENSES, GROUP_SETTLEMENTS, BUDGETS. Relationships shown: 10.

06. IOUs, recurring items and reminders

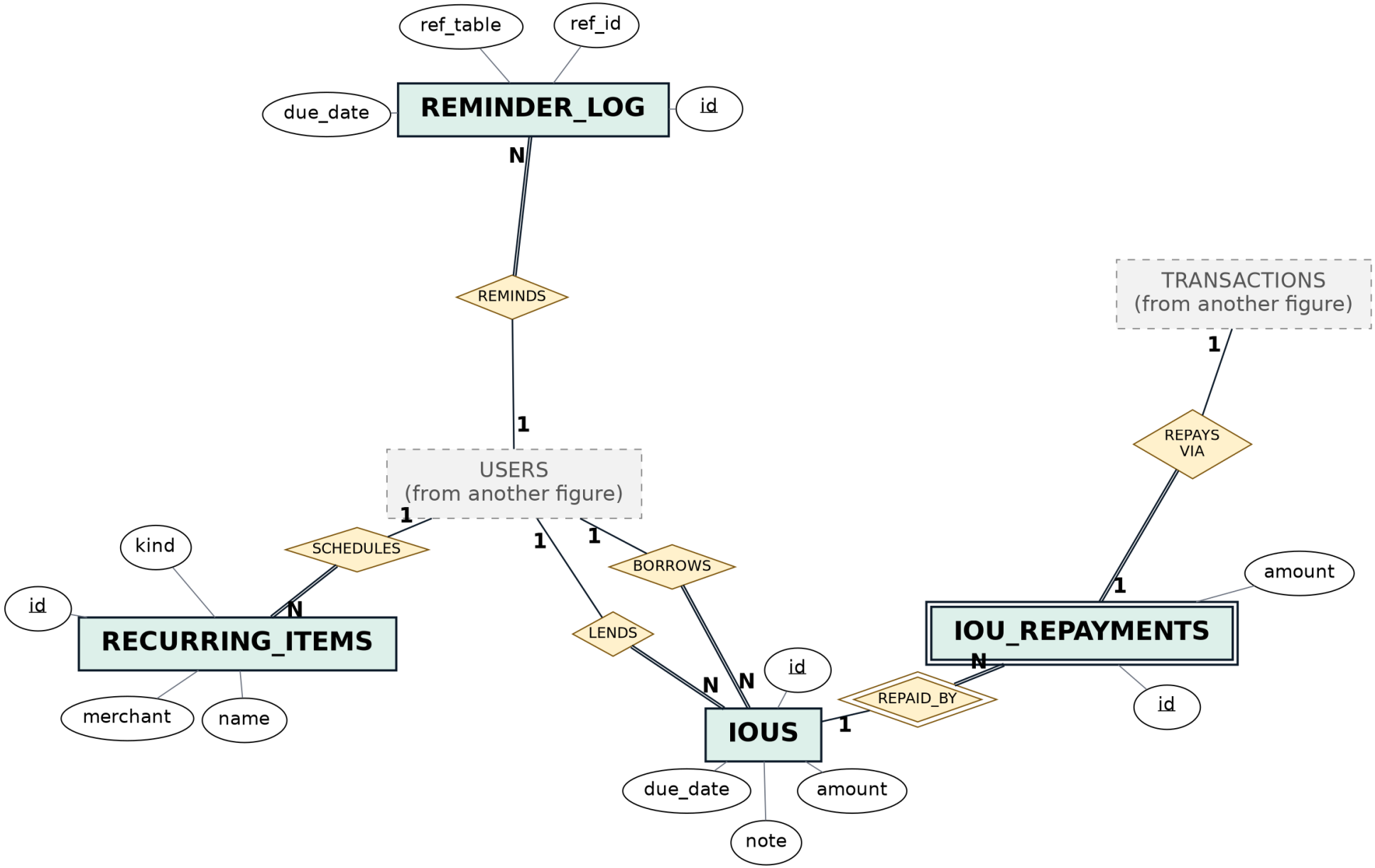


Figure 6. Entities: IOUS, IOU_REPAYMENTS, RECURRING_ITEMS, REMINDER_LOG. Relationships shown: 6.

07. Pockets and programmable money rules

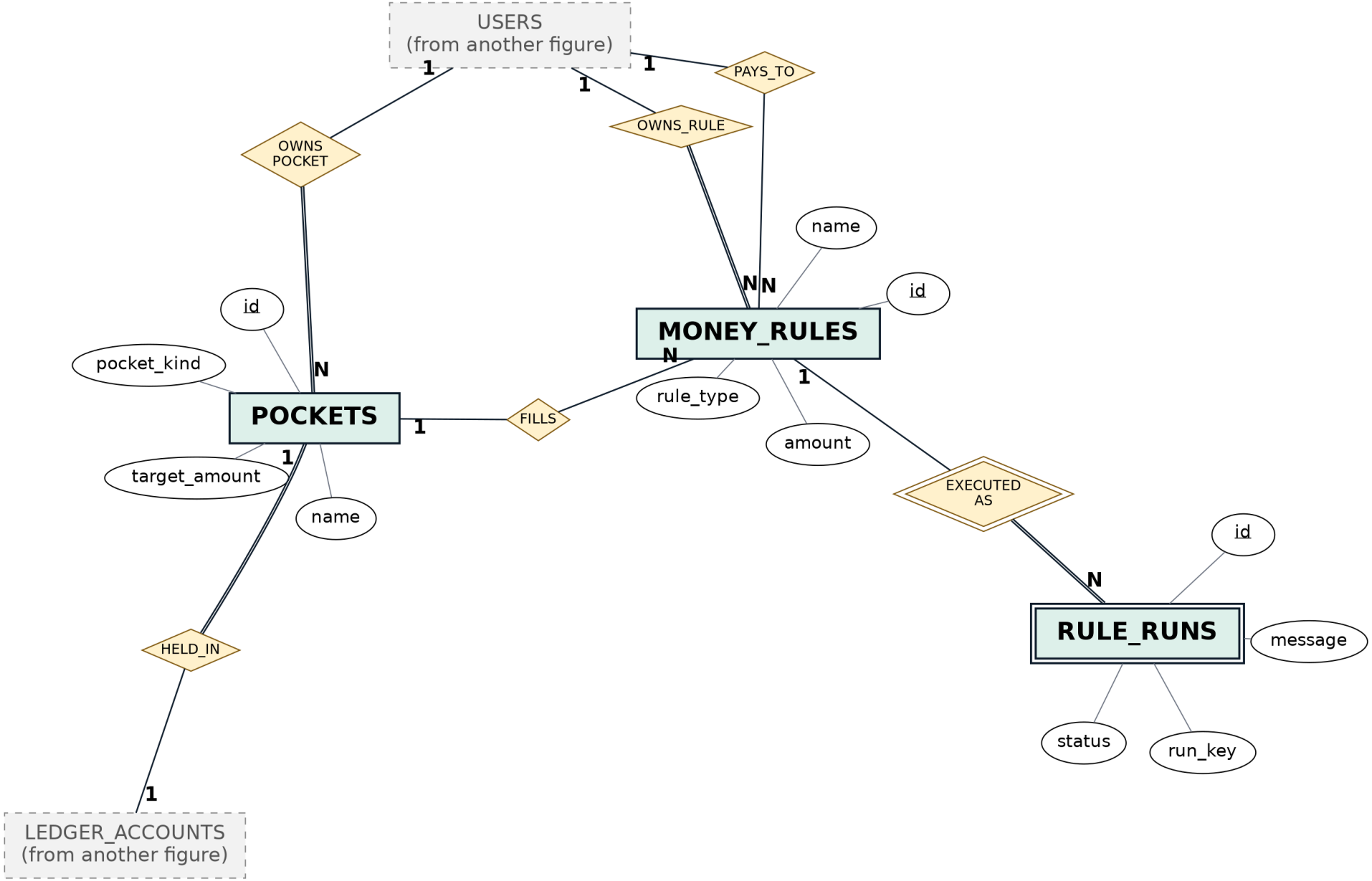


Figure 7. Entities: POCKETS, MONEY_RULES, RULE_RUNS. Relationships shown: 6.

08. Virtual tokens and offline QR

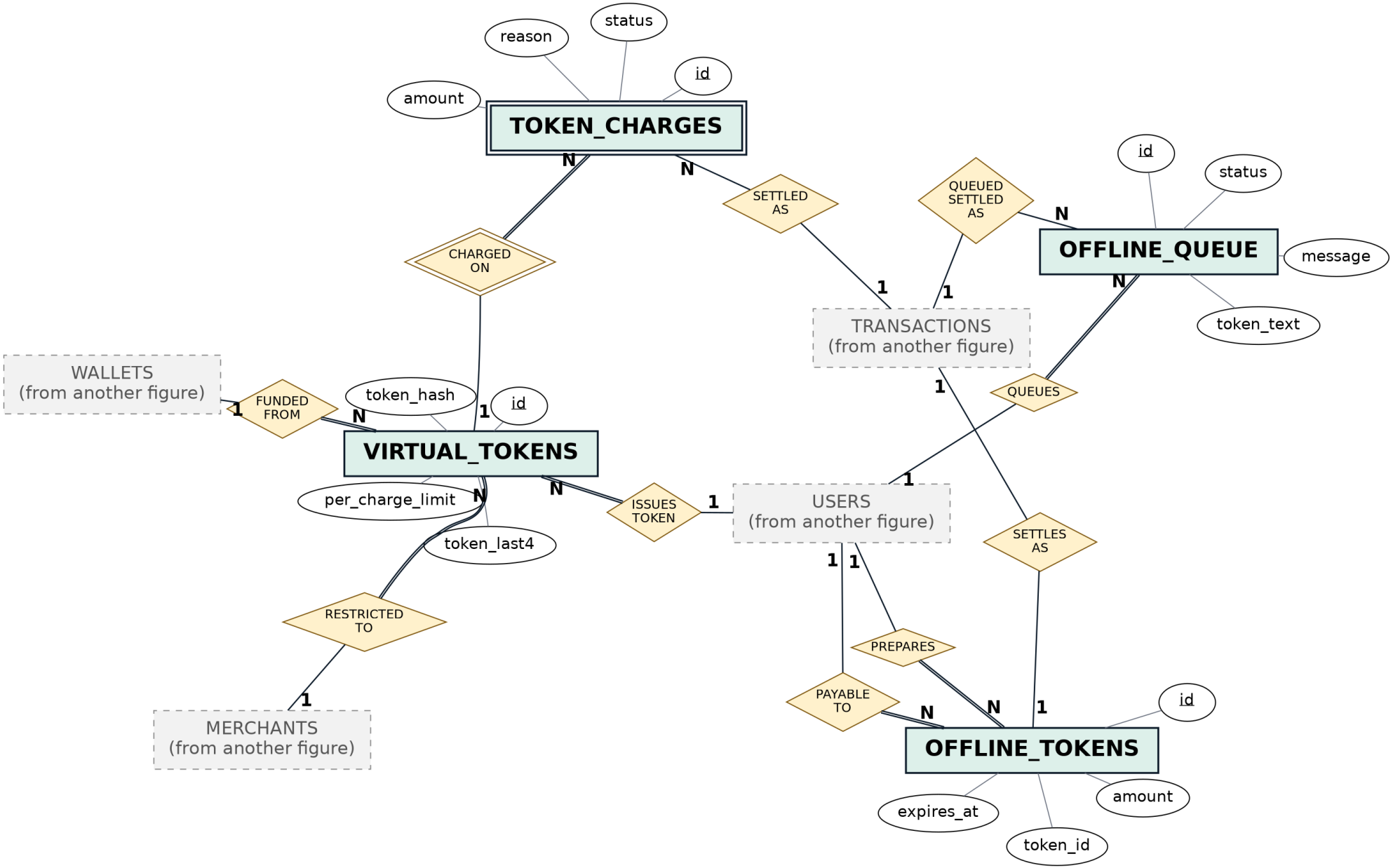


Figure 8. Entities: VIRTUAL_TOKENS, TOKEN_CHARGES, OFFLINE_TOKENS, OFFLINE_QUEUE. Relationships shown: 10.

09. Guardian controls and multi-approver authorization

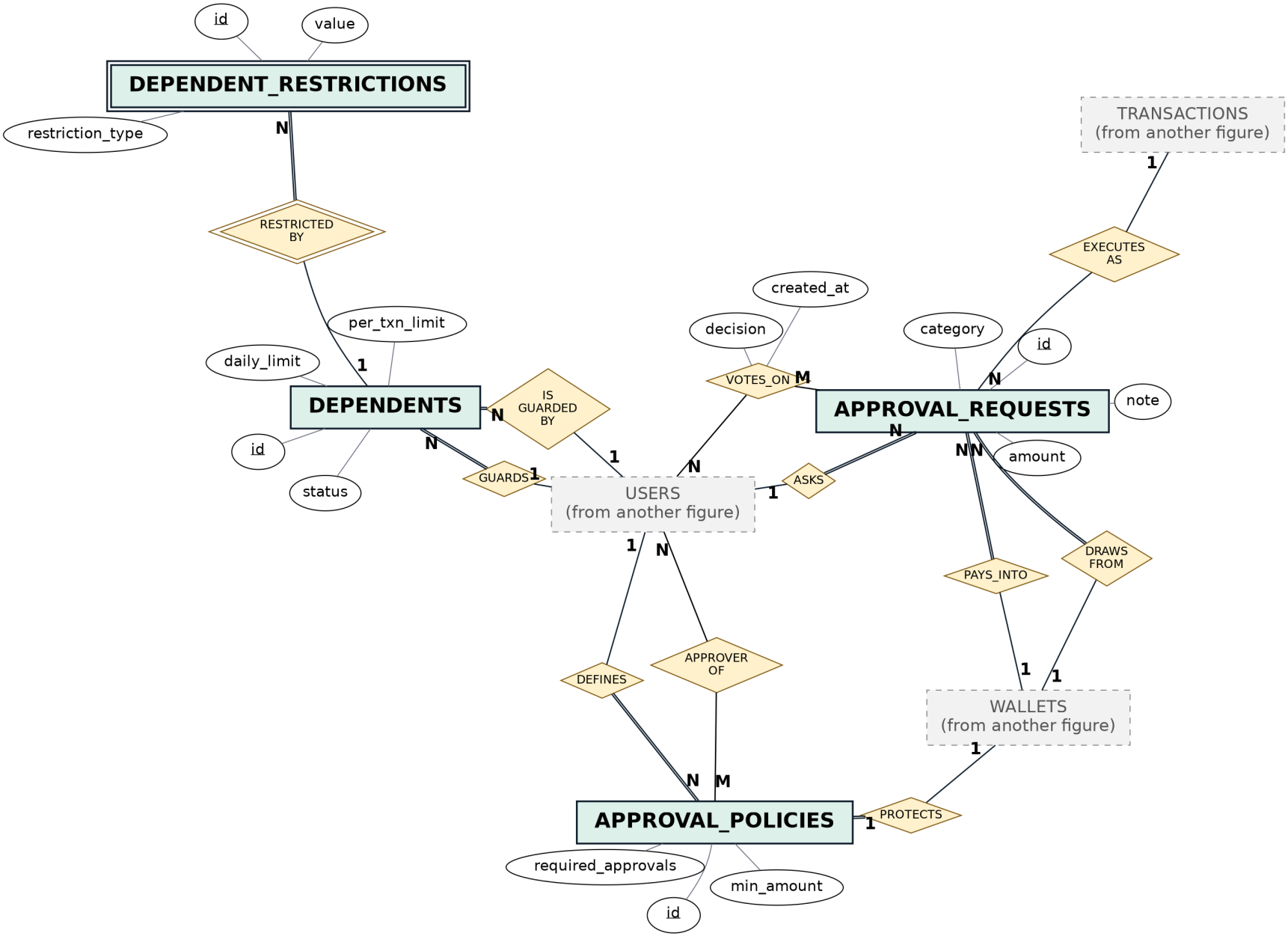


Figure 9. Entities: DEPENDENTS, DEPENDENT_RESTRICTIONS, APPROVAL_POLICIES, APPROVAL_REQUESTS. Relationships shown: 11.

10. Escrow, milestones and disputes

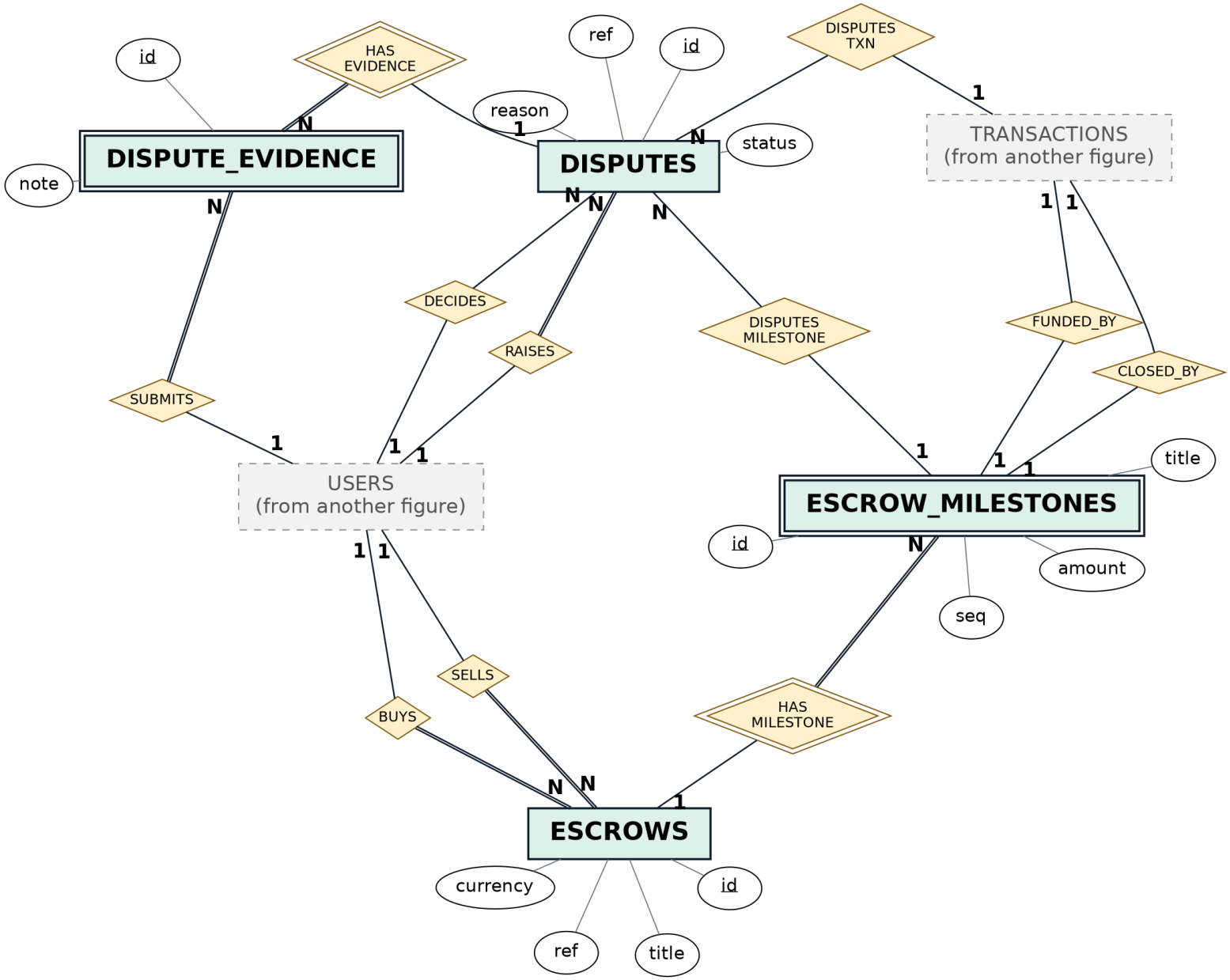


Figure 10. Entities: ESCROWS, ESCROW_MILESTONES, DISPUTES, DISPUTE_EVIDENCE. Relationships shown: 11.

11. Invoices and payments

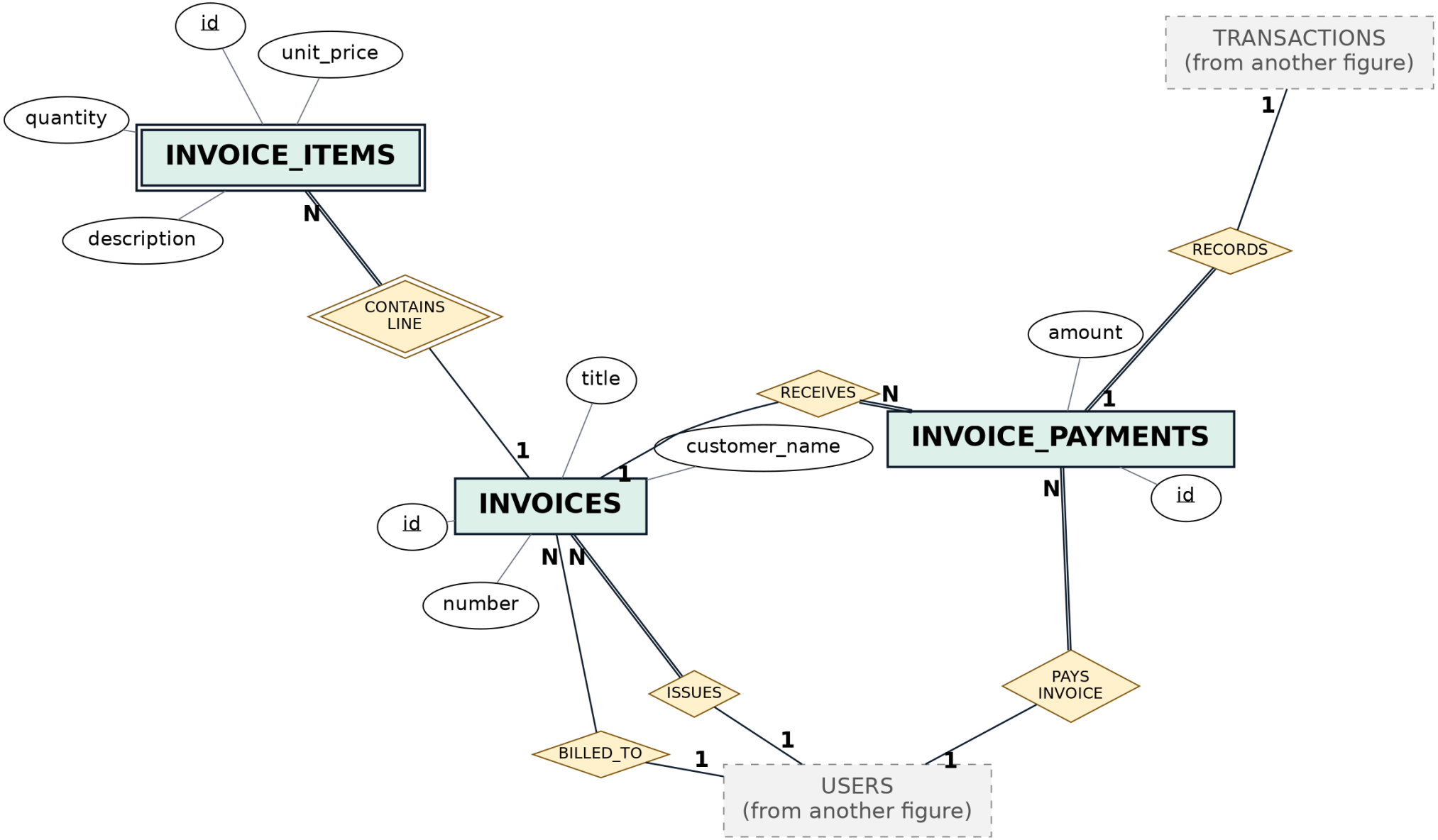


Figure 11. Entities: INVOICES, INVOICE_ITEMS, INVOICE_PAYMENTS. Relationships shown: 6.

3. List of entities and attributes

Each entity is listed with all of its attributes. Types are the MySQL/MariaDB types of the implemented schema. **PK** = primary key (underlined name), **FK** = foreign key (target in Section 5), **UQ** = unique, **NULL** = optional value.

Figure 01 - Identity, devices, tokens and idempotency

1. ROLES - Strong entity. Lookup of user roles (user, merchant, admin).

Attribute	Type	Key / constraint
id	tinyint unsigned	PK
name	varchar(20)	UQ

2. USERS - Strong entity. Accounts: credentials (hashed), role, status, lockout state.

Attribute	Type	Key / constraint
id	bigint unsigned	PK
name	varchar(100)	
email	varchar(150)	UQ
phone	varchar(20)	UQ, NULL
password_hash	varchar(255)	
pin_hash	varchar(255)	
role_id	tinyint unsigned	FK -> ROLES
status	enum('active', 'suspended')	
failed_logins	int	
locked_until	datetime	NULL
created_at	datetime	

3. USER_SETTINGS - Weak entity (identified by USERS through HAS_SETTINGS). Per-user language, privacy and notification preferences.

Attribute	Type	Key / constraint
user_id	bigint unsigned	PK, FK -> USERS
language	enum('en', 'bn')	
hide_balance	tinyint(1)	
notify_login	tinyint(1)	
notify_transaction	tinyint(1)	
notify_reminder	tinyint(1)	
sim_biometric	tinyint(1)	

4. DEVICES - Strong entity. Known devices/sessions per user; trust and revoke.

Attribute	Type	Key / constraint
id	bigint unsigned	PK
user_id	bigint unsigned	FK -> USERS
device_hash	char(64)	
label	varchar(150)	
ip_address	varchar(45)	
trusted	tinyint(1)	
revoked	tinyint(1)	
first_seen	datetime	
last_seen	datetime	

5. API_TOKENS - Strong entity. Hashed bearer tokens for the JSON API.

Attribute	Type	Key / constraint
id	bigint unsigned	PK
user_id	bigint unsigned	FK -> USERS
token_hash	char(64)	UQ
expires_at	datetime	
revoked	tinyint(1)	
created_at	datetime	

6. IDEMPOTENCY_KEYS - Strong entity. Unique (user,key,action) guard against duplicate money requests.

Attribute	Type	Key / constraint
id	bigint unsigned	PK
user_id	bigint unsigned	FK -> USERS
idem_key	varchar(64)	
action	varchar(40)	
txn_id	bigint unsigned	NULL
created_at	datetime	

Figure 02 - Notifications, risk alerts, audit, timelines and platform tables

7. NOTIFICATIONS - Strong entity. In-app notifications by category.

Attribute	Type	Key / constraint
id	bigint unsigned	PK
user_id	bigint unsigned	FK -> USERS
category	enum('login', 'transaction', 'reminder', 'security', 'system')	
title	varchar(150)	
body	varchar(400)	
is_read	tinyint(1)	
created_at	datetime	

8. RISK_ALERTS - Strong entity. Rule-based warnings (new recipient, velocity, ghost charge, replay...).

Attribute	Type	Key / constraint
id	bigint unsigned	PK
user_id	bigint unsigned	FK -> USERS
txn_id	bigint unsigned	NULL
alert_type	varchar(40)	
severity	enum('low', 'medium', 'high')	
message	varchar(255)	
status	enum('open', 'reviewed')	
created_at	datetime	

9. AUDIT_LOGS - Strong entity. Append-only record of sensitive actions with IP.

Attribute	Type	Key / constraint
id	bigint unsigned	PK
user_id	bigint unsigned	NULL
action	varchar(60)	
entity	varchar(40)	
entity_id	bigint unsigned	NULL
ip_address	varchar(45)	
details	varchar(400)	
created_at	datetime	

10. TIMELINE_EVENTS - Strong entity. Status timelines for escrow, disputes, invoices, transfers, approvals.

Attribute	Type	Key / constraint
id	bigint unsigned	PK
entity_type	enum('escrow', 'dispute', 'invoice', 'transfer', 'approval')	
entity_id	bigint unsigned	
actor_id	bigint unsigned	NULL
event	varchar(60)	
note	varchar(255)	
created_at	datetime	

11. RATE_LIMITS - Strong entity. Fixed-window counters for login, PIN, transfer and API limits.

Attribute	Type	Key / constraint
bucket	varchar(120)	PK
hits	int	
window_start	datetime	

12. SETTINGS - Strong entity. Configurable fees, limits, cancellation window.

Attribute	Type	Key / constraint
skey	varchar(60)	PK
svalue	varchar(255)	
description	varchar(200)	

Figure 03 - Ledger core: accounts, wallets, transactions, entries, holds

13. LEDGER_ACCOUNTS - Strong entity. Double-entry accounts (wallet, pocket, hold, escrow, fee, mint, fx, external) with cached balance.

Attribute	Type	Key / constraint
id	bigint unsigned	PK
code	varchar(40)	UQ
kind	enum('wallet', 'pocket', 'hold', 'escrow', 'fee', 'mint', 'fx', 'external')	
currency	char(3)	
name	varchar(100)	
balance	decimal(18, 2)	

14. WALLETS - Strong entity. One wallet per user and currency, bound to a ledger account.

Attribute	Type	Key / constraint
id	bigint unsigned	PK
user_id	bigint unsigned	FK -> USERS
currency	char(3)	
account_id	bigint unsigned	FK -> LEDGER_ACCOUNTS, UQ
frozen	tinyint(1)	
frozen_by	enum('none', 'user', 'admin')	
created_at	datetime	

15. TRANSACTIONS - Strong entity. Business-level transaction record and state.

Attribute	Type	Key / constraint
id	bigint unsigned	PK
ref	char(14)	UQ
type	enum('topup', 'transfer', 'request_payment', 'merchant_payment', 'settlement', 'iou_repayment', 'invoice_payment', 'bill_payment', 'refund', 'escrow_fund', 'escrow_release', 'escrow_refund', 'conversion', 'pocket_deposit', 'pocket_withdraw', 'rule_transfer', 'round_up', 'offline_payment', 'token_payment')	
status	enum('pending', 'held', 'completed', 'cancelled', 'failed', 'refunded', 'disputed')	
from_wallet_id	bigint unsigned	FK -> WALLETS, NULL
to_wallet_id	bigint unsigned	FK -> WALLETS, NULL
amount	decimal(18, 2)	
fee	decimal(18, 2)	
currency	char(3)	
category	varchar(30)	
note	varchar(255)	
created_by	bigint unsigned	FK -> USERS
refund_of	bigint unsigned	FK -> TRANSACTIONS, UQ, NULL
ref_table	varchar(30)	NULL
ref_id	bigint unsigned	NULL
created_at	datetime	
completed_at	datetime	NULL

16. LEDGER_ENTRIES - Strong entity. Immutable debit/credit lines; the source of truth for balances.

Attribute	Type	Key / constraint
id	bigint unsigned	PK
txn_id	bigint unsigned	FK -> TRANSACTIONS
phase	varchar(20)	
account_id	bigint unsigned	FK -> LEDGER_ACCOUNTS
direction	enum('D', 'C')	
amount	decimal(18, 2)	
currency	char(3)	
created_at	datetime	

17. TRANSFER_HOLDS - Weak entity (identified by TRANSACTIONS through HOLDS). Cancellation-window state for held transfers.

Attribute	Type	Key / constraint
id	bigint unsigned	PK
txn_id	bigint unsigned	FK -> TRANSACTIONS, UQ
release_at	datetime	
status	enum('held', 'released', 'cancelled')	
reason	varchar(120)	
resolved_at	datetime	NULL

Figure 04 - Payment requests, exchange rates, merchants, QR checkout

18. PAYMENT_REQUESTS - Strong entity. Money requests between users.

Attribute	Type	Key / constraint
id	bigint unsigned	PK
requester_id	bigint unsigned	FK -> USERS
payer_id	bigint unsigned	FK -> USERS
amount	decimal(18, 2)	
currency	char(3)	
note	varchar(255)	
status	enum('pending', 'paid', 'declined', 'cancelled')	
txn_id	bigint unsigned	FK -> TRANSACTIONS, NULL
created_at	datetime	

19. EXCHANGE_RATES - Strong entity. SIMULATED dated demonstration FX rates.

Attribute	Type	Key / constraint
id	int unsigned	PK
base_currency	char(3)	
quote_currency	char(3)	
rate	decimal(18, 6)	
as_of	date	
source	varchar(80)	

20. MERCHANTS - Strong entity. Merchant profile for a user.

Attribute	Type	Key / constraint
id	bigint unsigned	PK
user_id	bigint unsigned	FK -> USERS, UQ
business_name	varchar(120)	
category	varchar(30)	
status	enum('active', 'suspended')	

21. CHECKOUT_SESSIONS - Strong entity. Expiring QR checkout sessions.

Attribute	Type	Key / constraint
id	bigint unsigned	PK
merchant_id	bigint unsigned	FK -> MERCHANTS
ref	char(16)	UQ
amount	decimal(18, 2)	
note	varchar(200)	
status	enum('open', 'paid', 'expired', 'cancelled')	
expires_at	datetime	
paid_txn_id	bigint unsigned	FK -> TRANSACTIONS, UQ, NULL
created_at	datetime	

Figure 05 - Group finance and budgets

22. EXPENSE_GROUPS - Strong entity. Shared-expense groups.

Attribute	Type	Key / constraint
id	bigint unsigned	PK
name	varchar(100)	
group_type	enum('rent', 'trip', 'restaurant', 'groceries', 'subscription', 'other')	
created_by	bigint unsigned	FK -> USERS
created_at	datetime	

23. GROUP_EXPENSES - Strong entity. Expenses paid by a member.

Attribute	Type	Key / constraint
id	bigint unsigned	PK
group_id	bigint unsigned	FK -> EXPENSE_GROUPS
paid_by	bigint unsigned	FK -> USERS
description	varchar(150)	
category	varchar(30)	
amount	decimal(18, 2)	
split_type	enum('equal', 'custom', 'percent')	
expense_date	date	
created_at	datetime	

24. GROUP_SETTLEMENTS - Strong entity. Settlements linked to real wallet transfers.

Attribute	Type	Key / constraint
id	bigint unsigned	PK
group_id	bigint unsigned	FK -> EXPENSE_GROUPS
from_user	bigint unsigned	FK -> USERS
to_user	bigint unsigned	FK -> USERS
amount	decimal(18, 2)	
txn_id	bigint unsigned	FK -> TRANSACTIONS, UQ
created_at	datetime	

25. BUDGETS - Strong entity. Personal and shared monthly budgets (group_id 0 = personal).

Attribute	Type	Key / constraint
id	bigint unsigned	PK
user_id	bigint unsigned	FK -> USERS
group_id	bigint unsigned	
category	varchar(30)	
monthly_limit	decimal(18, 2)	
created_at	datetime	

Figure 06 - IOUs, recurring items and reminders

26. IOUS - Strong entity. Loans between users with due dates.

Attribute	Type	Key / constraint
id	bigint unsigned	PK
lender_id	bigint unsigned	FK -> USERS
borrower_id	bigint unsigned	FK -> USERS
amount	decimal(18, 2)	
note	varchar(200)	
due_date	date	
status	enum('open', 'partial', 'settled', 'cancelled')	
created_at	datetime	

27. IOU_REPAYMENTS - Weak entity (identified by IOUS through REPAID_BY). Partial repayments linked to transactions.

Attribute	Type	Key / constraint
id	bigint unsigned	PK
iou_id	bigint unsigned	FK -> IOUS
amount	decimal(18, 2)	
txn_id	bigint unsigned	FK -> TRANSACTIONS, UQ
created_at	datetime	

28. RECURRING_ITEMS - Strong entity. Bills, subscriptions, rent and known income.

Attribute	Type	Key / constraint
id	bigint unsigned	PK
user_id	bigint unsigned	FK -> USERS
kind	enum('income', 'bill', 'subscription', 'rent')	
name	varchar(100)	
merchant	varchar(100)	
amount	decimal(18, 2)	
currency	char(3)	
frequency	enum('weekly', 'monthly', 'yearly')	
next_due	date	
category	varchar(30)	
status	enum('active', 'paused', 'cancelled')	
created_at	datetime	

29. REMINDER_LOG - Strong entity. Ensures each reminder is sent once.

Attribute	Type	Key / constraint
id	bigint unsigned	PK
user_id	bigint unsigned	FK -> USERS
ref_table	varchar(30)	
ref_id	bigint unsigned	
due_date	date	
sent_at	datetime	

Figure 07 - Pockets and programmable money rules

30. POCKETS - Strong entity. Simulated goal pockets backed by pocket ledger accounts.

Attribute	Type	Key / constraint
id	bigint unsigned	PK
user_id	bigint unsigned	FK -> USERS
name	varchar(80)	
pocket_kind	enum('education', 'travel', 'emergency', 'monthly_expense', 'custom')	
target_amount	decimal(18, 2)	
target_date	date	NULL
account_id	bigint unsigned	FK -> LEDGER_ACCOUNTS, UQ
created_at	datetime	

31. MONEY_RULES - Strong entity. Programmable rules (scheduled, allocation, round-up, alert).

Attribute	Type	Key / constraint
id	bigint unsigned	PK
user_id	bigint unsigned	FK -> USERS
name	varchar(100)	
rule_type	enum('scheduled_transfer', 'income_allocation', 'round_up', 'low_balance_alert')	
target_user_id	bigint unsigned	FK -> USERS, NULL
target_pocket_id	bigint unsigned	FK -> POCKETS, NULL
amount	decimal(18, 2)	NULL
percent	decimal(5, 2)	NULL
round_to	int	NULL
threshold	decimal(18, 2)	NULL
frequency	enum('daily', 'weekly', 'monthly')	NULL
next_run	date	NULL
status	enum('active', 'paused', 'failed')	
fail_count	int	
last_message	varchar(200)	
created_at	datetime	

32. RULE_RUNS - Weak entity (identified by MONEY_RULES through EXECUTED_AS). Idempotent run log: unique (rule, run_key).

Attribute	Type	Key / constraint
id	bigint unsigned	PK
rule_id	bigint unsigned	FK -> MONEY_RULES
run_key	varchar(40)	
status	enum('success', 'failed', 'skipped')	
message	varchar(200)	
txn_id	bigint unsigned	NULL
ran_at	datetime	

Figure 08 - Virtual tokens and offline QR

33. VIRTUAL_TOKENS - Strong entity. Merchant-specific tokens (hashed) with limits.

Attribute	Type	Key / constraint
id	bigint unsigned	PK
user_id	bigint unsigned	FK -> USERS
wallet_id	bigint unsigned	FK -> WALLETS
merchant_id	bigint unsigned	FK -> MERCHANTS
token_hash	char(64)	UQ
token_last4	char(4)	
per_charge_limit	decimal(18, 2)	
monthly_limit	decimal(18, 2)	
status	enum('active', 'paused', 'cancelled')	
created_at	datetime	

34. TOKEN_CHARGES - Weak entity (identified by VIRTUAL_TOKENS through CHARGED_ON). Approved/declined merchant charges on tokens.

Attribute	Type	Key / constraint
id	bigint unsigned	PK
token_id	bigint unsigned	FK -> VIRTUAL_TOKENS
amount	decimal(18, 2)	
status	enum('approved', 'declined')	
reason	varchar(120)	
txn_id	bigint unsigned	FK -> TRANSACTIONS, NULL
created_at	datetime	

35. OFFLINE_TOKENS - Strong entity. One-time expiring offline payment tokens.

Attribute	Type	Key / constraint
id	bigint unsigned	PK
token_id	char(32)	UQ
user_id	bigint unsigned	FK -> USERS
payee_user_id	bigint unsigned	FK -> USERS
amount	decimal(18, 2)	
expires_at	datetime	
status	enum('issued', 'settled', 'rejected', 'expired')	
txn_id	bigint unsigned	FK -> TRANSACTIONS, UQ, NULL
created_at	datetime	

36. OFFLINE_QUEUE - Strong entity. Merchant offline queue awaiting sync.

Attribute	Type	Key / constraint
id	bigint unsigned	PK
merchant_user_id	bigint unsigned	FK -> USERS
token_text	text	
status	enum('queued', 'settled', 'rejected')	
message	varchar(200)	
txn_id	bigint unsigned	FK -> TRANSACTIONS, NULL
queued_at	datetime	
synced_at	datetime	NULL

Figure 09 - Guardian controls and multi-approver authorization

37. DEPENDENTS - Strong entity. Guardian-dependent link with spending limits.

Attribute	Type	Key / constraint
id	bigint unsigned	PK
guardian_id	bigint unsigned	FK -> USERS
dependent_id	bigint unsigned	FK -> USERS
per_txn_limit	decimal(18, 2)	
daily_limit	decimal(18, 2)	
status	enum('active', 'paused')	

38. DEPENDENT_RESTRICTIONS - Weak entity (identified by DEPENDENTS through RESTRICTED_BY). Blocked categories or merchants.

Attribute	Type	Key / constraint
id	bigint unsigned	PK
dependent_row_id	bigint unsigned	FK -> DEPENDENTS
restriction_type	enum('category', 'merchant')	
value	varchar(80)	

39. APPROVAL_POLICIES - Strong entity. Threshold and required approvals per wallet.

Attribute	Type	Key / constraint
id	bigint unsigned	PK
wallet_id	bigint unsigned	FK -> WALLETS, UQ
min_amount	decimal(18, 2)	
required_approvals	int	
created_by	bigint unsigned	FK -> USERS

40. APPROVAL_REQUESTS - Strong entity. Payments waiting for approval.

Attribute	Type	Key / constraint
id	bigint unsigned	PK
wallet_id	bigint unsigned	FK -> WALLETS
requested_by	bigint unsigned	FK -> USERS
to_wallet_id	bigint unsigned	FK -> WALLETS
amount	decimal(18, 2)	
category	varchar(30)	
note	varchar(255)	
status	enum('pending', 'executed', 'rejected', 'failed')	
txn_id	bigint unsigned	FK -> TRANSACTIONS, NULL
created_at	datetime	

Figure 10 - Escrow, milestones and disputes

41. ESCROWS - Strong entity. Escrow contracts between buyer and seller.

Attribute	Type	Key / constraint
id	bigint unsigned	PK
ref	char(12)	UQ
buyer_id	bigint unsigned	FK -> USERS
seller_id	bigint unsigned	FK -> USERS
title	varchar(150)	
currency	char(3)	
status	enum('open', 'completed', 'disputed', 'cancelled')	
created_at	datetime	

42. ESCROW_MILESTONES - Weak entity (identified by ESCROWS through HAS_MILESTONE). Milestone amounts and funding/closing transactions.

Attribute	Type	Key / constraint
id	bigint unsigned	PK
escrow_id	bigint unsigned	FK -> ESCROWS
seq	int	
title	varchar(150)	
amount	decimal(18, 2)	
due_date	date	NULL
status	enum('pending', 'funded', 'submitted', 'released', 'refunded', 'disputed')	
funded_txn_id	bigint unsigned	FK -> TRANSACTIONS, UQ, NULL
closing_txn_id	bigint unsigned	FK -> TRANSACTIONS, UQ, NULL

43. DISPUTES - Strong entity. Dispute cases and admin decisions.

Attribute	Type	Key / constraint
id	bigint unsigned	PK
ref	char(10)	UQ
raised_by	bigint unsigned	FK -> USERS
txn_id	bigint unsigned	FK -> TRANSACTIONS, NULL
milestone_id	bigint unsigned	FK -> ESCROW_MILESTONES, NULL
reason	varchar(255)	
status	enum('open', 'under_review', 'resolved', 'rejected')	
resolution	enum('none', 'refund', 'release')	
admin_id	bigint unsigned	FK -> USERS, NULL
decision_note	varchar(255)	
created_at	datetime	
resolved_at	datetime	NULL

44. DISPUTE_EVIDENCE - Weak entity (identified by DISPUTES through HAS_EVIDENCE). Evidence notes.

Attribute	Type	Key / constraint
id	bigint unsigned	PK
dispute_id	bigint unsigned	FK -> DISPUTES
user_id	bigint unsigned	FK -> USERS
note	varchar(400)	
created_at	datetime	

Figure 11 - Invoices and payments

45. INVOICES - Strong entity. Invoices with shareable link token.

Attribute	Type	Key / constraint
id	bigint unsigned	PK
number	varchar(20)	UQ
issuer_id	bigint unsigned	FK -> USERS
customer_id	bigint unsigned	FK -> USERS, NULL
customer_name	varchar(100)	
title	varchar(150)	
currency	char(3)	
due_date	date	
status	enum('draft', 'sent', 'partial', 'paid', 'overdue', 'cancelled')	
link_token	char(32)	UQ
created_at	datetime	

46. INVOICE_ITEMS - Weak entity (identified by INVOICES through CONTAINS_LINE). Invoice line items.

Attribute	Type	Key / constraint
id	bigint unsigned	PK
invoice_id	bigint unsigned	FK -> INVOICES
description	varchar(150)	
quantity	decimal(10, 2)	
unit_price	decimal(18, 2)	

47. INVOICE_PAYMENTS - Strong entity. Payments reconciled to invoices.

Attribute	Type	Key / constraint
id	bigint unsigned	PK
invoice_id	bigint unsigned	FK -> INVOICES
txn_id	bigint unsigned	FK -> TRANSACTIONS, UQ
payer_id	bigint unsigned	FK -> USERS
amount	decimal(18, 2)	
created_at	datetime	

Many-to-many relationships and their attributes

A many-to-many relationship cannot be stored directly; each is mapped to a link relation that holds the two foreign keys and the attributes of the relationship.

Relationship	Entities	Relationship attributes	Link relation
MEMBER_OF	EXPENSE_GROUPS (M) - USERS (N)	joined_at	GROUP_MEMBERS
SPLIT_AMONG	GROUP_EXPENSES (M) - USERS (N)	share_amount, share_percent	EXPENSE_SPLITS
APPROVER_OF	APPROVAL_POLICIES (M) - USERS (N)	(none)	APPROVAL_POLICY_APPROVERS
VOTES_ON	APPROVAL_REQUESTS (M) - USERS (N)	decision, created_at	APPROVAL_VOTES

4. Relationships and cardinality

Every relationship of the ERD with its cardinality and participation. **Participation**: total means every instance of the entity must take part (the foreign key is NOT NULL); partial means it may be absent. The parent side of a one-to-many relationship is always partial in this design, so only the child side is listed. Entity A is the parent (the 1 side).

#	Relationship	Entity A (card.)	Entity B (card.)	Type	Child part.	Implemented by (FK column)
1	OWNS_TOKEN	USERS (1)	API_TOKENS (N)	1:N	total	user_id
2	USES_DEVICE	USERS (1)	DEVICES (N)	1:N	total	user_id
3	SUBMITS_KEY	USERS (1)	IDEMPOTENCY_KEYS (N)	1:N	total	user_id
4	HAS_SETTINGS (identifying)	USERS (1)	USER_SETTINGS (N)	1:N	total	user_id
5	HAS_ROLE	ROLES (1)	USERS (N)	1:N	total	role_id
6	NOTIFIES	USERS (1)	NOTIFICATIONS (N)	1:N	total	user_id
7	RAISED_FOR	USERS (1)	RISK_ALERTS (N)	1:N	total	user_id
8	AFFECTS	LEDGER_ACCOUNTS (1)	LEDGER_ENTRIES (N)	1:N	total	account_id
9	POSTS	TRANSACTIONS (1)	LEDGER_ENTRIES (N)	1:N	total	txn_id
10	INITIATES	USERS (1)	TRANSACTIONS (N)	1:N	total	created_by
11	DEBITS	WALLETS (1)	TRANSACTIONS (N)	1:N	partial	from_wallet_id
12	REFUNDS	TRANSACTIONS (1)	TRANSACTIONS (1)	1:1	partial	refund_of
13	CREDITS	WALLETS (1)	TRANSACTIONS (N)	1:N	partial	to_wallet_id
14	HOLDS (identifying)	TRANSACTIONS (1)	TRANSFER_HOLDS (1)	1:1	total	txn_id
15	BACKED_BY	LEDGER_ACCOUNTS (1)	WALLETS (1)	1:1	total	account_id
16	OWNS_WALLET	USERS (1)	WALLETS (N)	1:N	total	user_id
17	OFFERS	MERCHANTS (1)	CHECKOUT_SESSIONS (N)	1:N	total	merchant_id
18	PAID_BY	TRANSACTIONS (1)	CHECKOUT_SESSIONS (1)	1:1	partial	paid_txn_id
19	RUNS	USERS (1)	MERCHANTS (1)	1:1	total	user_id
20	IS_ASKED	USERS (1)	PAYMENT_REQUESTS (N)	1:N	total	payer_id
21	REQUESTS	USERS (1)	PAYMENT_REQUESTS (N)	1:N	total	requester_id
22	SETTLED_BY	TRANSACTIONS (1)	PAYMENT_REQUESTS (N)	1:N	partial	txn_id
23	SETS	USERS (1)	BUDGETS (N)	1:N	total	user_id
24	CREATES	USERS (1)	EXPENSE_GROUPS (N)	1:N	total	created_by
25	CONTAINS_EXPENSE	EXPENSE_GROUPS (1)	GROUP_EXPENSES (N)	1:N	total	group_id
26	PAYS	USERS (1)	GROUP_EXPENSES (N)	1:N	total	paid_by
27	SETTLES_FROM	USERS (1)	GROUP_SETTLEMENTS (N)	1:N	total	from_user
28	SETTLES_IN	EXPENSE_GROUPS (1)	GROUP_SETTLEMENTS (N)	1:N	total	group_id
29	SETTLES_TO	USERS (1)	GROUP_SETTLEMENTS (N)	1:N	total	to_user
30	SETTLED_VIA	TRANSACTIONS (1)	GROUP_SETTLEMENTS (1)	1:1	total	txn_id
31	MEMBER_OF	EXPENSE_GROUPS (M)	USERS (N)	M:N	both partial	link relation group_members
32	SPLIT_AMONG	GROUP_EXPENSES (M)	USERS (N)	M:N	both partial	link relation expense_splits
33	REPAID_BY (identifying)	IOUS (1)	IOU_REPAYMENTS (N)	1:N	total	iou_id
34	REPAYS_VIA	TRANSACTIONS (1)	IOU_REPAYMENTS (1)	1:1	total	txn_id
35	BORROWS	USERS (1)	IOUS (N)	1:N	total	borrower_id
36	LENDS	USERS (1)	IOUS (N)	1:N	total	lender_id
37	SCHEDULES	USERS (1)	RECURRING_ITEMS (N)	1:N	total	user_id
38	REMINDS	USERS (1)	REMINDER_LOG (N)	1:N	total	user_id
39	FILLS	POCKETS (1)	MONEY_RULES (N)	1:N	partial	target_pocket_id
40	PAYS_TO	USERS (1)	MONEY_RULES (N)	1:N	partial	target_user_id
41	OWNS_RULE	USERS (1)	MONEY_RULES (N)	1:N	total	user_id
42	HELD_IN	LEDGER_ACCOUNTS (1)	POCKETS (1)	1:1	total	account_id
43	OWNS_POCKET	USERS (1)	POCKETS (N)	1:N	total	user_id
44	EXECUTED_AS (identifying)	MONEY_RULES (1)	RULE_RUNS (N)	1:N	total	rule_id
45	QUEUED_SETTLED_AS	TRANSACTIONS (1)	OFFLINE_QUEUE (N)	1:N	partial	txn_id
46	QUEUES	USERS (1)	OFFLINE_QUEUE (N)	1:N	total	merchant_user_id

#	Relationship	Entity A (card.)	Entity B (card.)	Type	Child part.	Implemented by (FK column)
47	PAYABLE_TO	USERS (1)	OFFLINE_TOKENS (N)	1:N	total	payee_user_id
48	SETTLES_AS	TRANSACTIONS (1)	OFFLINE_TOKENS (1)	1:1	partial	txn_id
49	PREPARES	USERS (1)	OFFLINE_TOKENS (N)	1:N	total	user_id
50	CHARGED_ON (identifying)	VIRTUAL_TOKENS (1)	TOKEN_CHARGES (N)	1:N	total	token_id
51	SETTLED_AS	TRANSACTIONS (1)	TOKEN_CHARGES (N)	1:N	partial	txn_id
52	RESTRICTED_TO	MERCHANTS (1)	VIRTUAL_TOKENS (N)	1:N	total	merchant_id
53	ISSUES_TOKEN	USERS (1)	VIRTUAL_TOKENS (N)	1:N	total	user_id
54	FUNDED_FROM	WALLETS (1)	VIRTUAL_TOKENS (N)	1:N	total	wallet_id
55	DEFINES	USERS (1)	APPROVAL_POLICIES (N)	1:N	total	created_by
56	PROTECTS	WALLETS (1)	APPROVAL_POLICIES (1)	1:1	total	wallet_id
57	PAYS_INTO	WALLETS (1)	APPROVAL_REQUESTS (N)	1:N	total	to_wallet_id
58	EXECUTES_AS	TRANSACTIONS (1)	APPROVAL_REQUESTS (N)	1:N	partial	txn_id
59	ASKS	USERS (1)	APPROVAL_REQUESTS (N)	1:N	total	requested_by
60	DRAWS_FROM	WALLETS (1)	APPROVAL_REQUESTS (N)	1:N	total	wallet_id
61	RESTRICTED_BY (identifying)	DEPENDENTS (1)	DEPENDENT_RESTRICTIONS (N)	1:N	total	dependent_row_id
62	IS_GUARDED_BY	USERS (1)	DEPENDENTS (N)	1:N	total	dependent_id
63	GUARDS	USERS (1)	DEPENDENTS (N)	1:N	total	guardian_id
64	APPROVER_OF	APPROVAL_POLICIES (M)	USERS (N)	M:N	both partial	link relation approval_policy_approvers
65	VOTES_ON	APPROVAL_REQUESTS (M)	USERS (N)	M:N	both partial	link relation approval_votes
66	HAS_EVIDENCE (identifying)	DISPUTES (1)	DISPUTE_EVIDENCE (N)	1:N	total	dispute_id
67	SUBMITS	USERS (1)	DISPUTE_EVIDENCE (N)	1:N	total	user_id
68	DECIDES	USERS (1)	DISPUTES (N)	1:N	partial	admin_id
69	DISPUTES_MILESTONE	ESCROW_MILESTONES (1)	DISPUTES (N)	1:N	partial	milestone_id
70	DISPUTES_TXN	TRANSACTIONS (1)	DISPUTES (N)	1:N	partial	txn_id
71	RAISES	USERS (1)	DISPUTES (N)	1:N	total	raised_by
72	CLOSED_BY	TRANSACTIONS (1)	ESCROW_MILESTONES (1)	1:1	partial	closing_txn_id
73	HAS_MILESTONE (identifying)	ESCROWS (1)	ESCROW_MILESTONES (N)	1:N	total	escrow_id
74	FUNDED_BY	TRANSACTIONS (1)	ESCROW_MILESTONES (1)	1:1	partial	funded_txn_id
75	BUYS	USERS (1)	ESCROWS (N)	1:N	total	buyer_id
76	SELLS	USERS (1)	ESCROWS (N)	1:N	total	seller_id
77	CONTAINS_LINE (identifying)	INVOICES (1)	INVOICE_ITEMS (N)	1:N	total	invoice_id
78	RECEIVES	INVOICES (1)	INVOICE_PAYMENTS (N)	1:N	total	invoice_id
79	RECORDS	TRANSACTIONS (1)	INVOICE_PAYMENTS (1)	1:1	total	txn_id
80	PAYS_INVOICE	USERS (1)	INVOICE_PAYMENTS (N)	1:N	total	payer_id
81	BILLED_TO	USERS (1)	INVOICES (N)	1:N	partial	customer_id
82	ISSUES	USERS (1)	INVOICES (N)	1:N	total	issuer_id

5. Foreign key constraints

All 86 foreign keys of the schema. **ON DELETE** shows the referential action; RESTRICT is used for almost all financial tables so that history can never be deleted by accident. A foreign key marked **1:1** also carries a UNIQUE constraint.

#	Child relation	FK column	References (parent.column)	Null ?	Card.	ON DELETE
1	API_TOKENS	user_id	USERS.id	no	N:1	CASCADE
2	APPROVAL_POLICIES	created_by	USERS.id	no	N:1	RESTRICT
3	APPROVAL_POLICIES	wallet_id	WALLETS.id	no	1:1	RESTRICT
4	APPROVAL_POLICY_APPROVERS	policy_id	APPROVAL_POLICIES.id	no	N:1	CASCADE
5	APPROVAL_POLICY_APPROVERS	user_id	USERS.id	no	N:1	RESTRICT
6	APPROVAL_REQUESTS	requested_by	USERS.id	no	N:1	RESTRICT
7	APPROVAL_REQUESTS	to_wallet_id	WALLETS.id	no	N:1	RESTRICT
8	APPROVAL_REQUESTS	txn_id	TRANSACTIONS.id	yes	N:1	RESTRICT
9	APPROVAL_REQUESTS	wallet_id	WALLETS.id	no	N:1	RESTRICT
10	APPROVAL_VOTES	approver_id	USERS.id	no	N:1	RESTRICT
11	APPROVAL_VOTES	request_id	APPROVAL_REQUESTS.id	no	N:1	CASCADE
12	BUDGETS	user_id	USERS.id	no	N:1	RESTRICT
13	CHECKOUT_SESSIONS	merchant_id	MERCHANTS.id	no	N:1	RESTRICT
14	CHECKOUT_SESSIONS	paid_txn_id	TRANSACTIONS.id	yes	1:1	RESTRICT
15	DEPENDENT_RESTRICTIONS	dependent_row_id	DEPENDENTS.id	no	N:1	CASCADE
16	DEPENDENTS	dependent_id	USERS.id	no	N:1	RESTRICT
17	DEPENDENTS	guardian_id	USERS.id	no	N:1	RESTRICT
18	DEVICES	user_id	USERS.id	no	N:1	CASCADE
19	DISPUTE_EVIDENCE	dispute_id	DISPUTES.id	no	N:1	CASCADE
20	DISPUTE_EVIDENCE	user_id	USERS.id	no	N:1	RESTRICT
21	DISPUTES	admin_id	USERS.id	yes	N:1	RESTRICT
22	DISPUTES	milestone_id	ESCROW_MILESTONES.id	yes	N:1	RESTRICT
23	DISPUTES	raised_by	USERS.id	no	N:1	RESTRICT
24	DISPUTES	txn_id	TRANSACTIONS.id	yes	N:1	RESTRICT
25	ESCROW_MILESTONES	closing_txn_id	TRANSACTIONS.id	yes	1:1	RESTRICT
26	ESCROW_MILESTONES	escrow_id	ESCROWS.id	no	N:1	CASCADE
27	ESCROW_MILESTONES	funded_txn_id	TRANSACTIONS.id	yes	1:1	RESTRICT
28	ESCROWS	buyer_id	USERS.id	no	N:1	RESTRICT
29	ESCROWS	seller_id	USERS.id	no	N:1	RESTRICT
30	EXPENSE_GROUPS	created_by	USERS.id	no	N:1	RESTRICT
31	EXPENSE_SPLITS	expense_id	GROUP_EXPENSES.id	no	N:1	CASCADE
32	EXPENSE_SPLITS	user_id	USERS.id	no	N:1	RESTRICT
33	GROUP_EXPENSES	group_id	EXPENSE_GROUPS.id	no	N:1	CASCADE
34	GROUP_EXPENSES	paid_by	USERS.id	no	N:1	RESTRICT
35	GROUP_MEMBERS	group_id	EXPENSE_GROUPS.id	no	N:1	CASCADE
36	GROUP_MEMBERS	user_id	USERS.id	no	N:1	RESTRICT
37	GROUP_SETTLEMENTS	from_user	USERS.id	no	N:1	RESTRICT
38	GROUP_SETTLEMENTS	group_id	EXPENSE_GROUPS.id	no	N:1	RESTRICT
39	GROUP_SETTLEMENTS	to_user	USERS.id	no	N:1	RESTRICT
40	GROUP_SETTLEMENTS	txn_id	TRANSACTIONS.id	no	1:1	RESTRICT
41	IDEMPOTENCY_KEYS	user_id	USERS.id	no	N:1	RESTRICT
42	INVOICE_ITEMS	invoice_id	INVOICES.id	no	N:1	CASCADE
43	INVOICE_PAYMENTS	invoice_id	INVOICES.id	no	N:1	RESTRICT
44	INVOICE_PAYMENTS	payer_id	USERS.id	no	N:1	RESTRICT

#	Child relation	FK column	References (parent.column)	Null ?	Card.	ON DELETE
45	INVOICE_PAYMENTS	txn_id	TRANSACTIONS.id	no	1:1	RESTRICT
46	INVOICES	customer_id	USERS.id	yes	N:1	RESTRICT
47	INVOICES	issuer_id	USERS.id	no	N:1	RESTRICT
48	IOU_REPAYMENTS	iou_id	IOUS.id	no	N:1	RESTRICT
49	IOU_REPAYMENTS	txn_id	TRANSACTIONS.id	no	1:1	RESTRICT
50	IOUS	borrower_id	USERS.id	no	N:1	RESTRICT
51	IOUS	lender_id	USERS.id	no	N:1	RESTRICT
52	LEDGER_ENTRIES	account_id	LEDGER_ACCOUNTS.id	no	N:1	RESTRICT
53	LEDGER_ENTRIES	txn_id	TRANSACTIONS.id	no	N:1	RESTRICT
54	MERCHANTS	user_id	USERS.id	no	1:1	RESTRICT
55	MONEY_RULES	target_pocket_id	POCKETS.id	yes	N:1	RESTRICT
56	MONEY_RULES	target_user_id	USERS.id	yes	N:1	RESTRICT
57	MONEY_RULES	user_id	USERS.id	no	N:1	RESTRICT
58	NOTIFICATIONS	user_id	USERS.id	no	N:1	CASCADE
59	OFFLINE_QUEUE	merchant_user_id	USERS.id	no	N:1	RESTRICT
60	OFFLINE_QUEUE	txn_id	TRANSACTIONS.id	yes	N:1	RESTRICT
61	OFFLINE_TOKENS	payee_user_id	USERS.id	no	N:1	RESTRICT
62	OFFLINE_TOKENS	txn_id	TRANSACTIONS.id	yes	1:1	RESTRICT
63	OFFLINE_TOKENS	user_id	USERS.id	no	N:1	RESTRICT
64	PAYMENT_REQUESTS	payer_id	USERS.id	no	N:1	RESTRICT
65	PAYMENT_REQUESTS	requester_id	USERS.id	no	N:1	RESTRICT
66	PAYMENT_REQUESTS	txn_id	TRANSACTIONS.id	yes	N:1	RESTRICT
67	POCKETS	account_id	LEDGER_ACCOUNTS.id	no	1:1	RESTRICT
68	POCKETS	user_id	USERS.id	no	N:1	RESTRICT
69	RECURRING_ITEMS	user_id	USERS.id	no	N:1	RESTRICT
70	REMINDER_LOG	user_id	USERS.id	no	N:1	RESTRICT
71	RISK_ALERTS	user_id	USERS.id	no	N:1	RESTRICT
72	RULE_RUNS	rule_id	MONEY_RULES.id	no	N:1	CASCADE
73	TOKEN_CHARGES	token_id	VIRTUAL_TOKENS.id	no	N:1	RESTRICT
74	TOKEN_CHARGES	txn_id	TRANSACTIONS.id	yes	N:1	RESTRICT
75	TRANSACTIONS	created_by	USERS.id	no	N:1	RESTRICT
76	TRANSACTIONS	from_wallet_id	WALLETS.id	yes	N:1	RESTRICT
77	TRANSACTIONS	refund_of	TRANSACTIONS.id	yes	1:1	RESTRICT
78	TRANSACTIONS	to_wallet_id	WALLETS.id	yes	N:1	RESTRICT
79	TRANSFER_HOLDS	txn_id	TRANSACTIONS.id	no	1:1	RESTRICT
80	USER_SETTINGS	user_id	USERS.id	no	N:1	CASCADE
81	USERS	role_id	ROLES.id	no	N:1	RESTRICT
82	VIRTUAL_TOKENS	merchant_id	MERCHANTS.id	no	N:1	RESTRICT
83	VIRTUAL_TOKENS	user_id	USERS.id	no	N:1	RESTRICT
84	VIRTUAL_TOKENS	wallet_id	WALLETS.id	no	N:1	RESTRICT
85	WALLETS	account_id	LEDGER_ACCOUNTS.id	no	1:1	RESTRICT
86	WALLETS	user_id	USERS.id	no	N:1	RESTRICT

6. Logical schema

Format: **RELATION (attribute [PK], attribute, attribute [FK → OtherRelation.column])**. [PK] primary key, [FK → ...] foreign key, [UQ] unique. Relation names are the capitalised table names.

API_TOKENS (id [PK], user_id [FK → USERS.id], token_hash [UQ], expires_at, revoked, created_at)

APPROVAL_POLICIES (id [PK], wallet_id [FK → WALLETS.id, UQ], min_amount, required_approvals, created_by [FK → USERS.id])

APPROVAL_POLICY_APPROVERS (policy_id [PK, FK → APPROVAL_POLICIES.id], user_id [PK, FK → USERS.id]) (link relation for APPROVER_OF)

APPROVAL_REQUESTS (id [PK], wallet_id [FK → WALLETS.id], requested_by [FK → USERS.id], to_wallet_id [FK → WALLETS.id], amount, category, note, status, txn_id [FK → TRANSACTIONS.id], created_at)

APPROVAL_VOTES (id [PK], request_id [FK → APPROVAL_REQUESTS.id], approver_id [FK → USERS.id], decision, created_at) (link relation for VOTES_ON)

AUDIT_LOGS (id [PK], user_id, action, entity, entity_id, ip_address, details, created_at)

BUDGETS (id [PK], user_id [FK → USERS.id], group_id, category, monthly_limit, created_at)

CHECKOUT_SESSIONS (id [PK], merchant_id [FK → MERCHANTS.id], ref [UQ], amount, note, status, expires_at, paid_txn_id [FK → TRANSACTIONS.id, UQ], created_at)

DEPENDENT_RESTRICTIONS (id [PK], dependent_row_id [FK → DEPENDENTS.id], restriction_type, value) (weak entity)

DEPENDENTS (id [PK], guardian_id [FK → USERS.id], dependent_id [FK → USERS.id], per_txn_limit, daily_limit, status)

DEVICES (id [PK], user_id [FK → USERS.id], device_hash, label, ip_address, trusted, revoked, first_seen, last_seen)

DISPUTE_EVIDENCE (id [PK], dispute_id [FK → DISPUTES.id], user_id [FK → USERS.id], note, created_at) (weak entity)

DISPUTES (id [PK], ref [UQ], raised_by [FK → USERS.id], txn_id [FK → TRANSACTIONS.id], milestone_id [FK → ESCROW_MILESTONES.id], reason, status, resolution, admin_id [FK → USERS.id], decision_note, created_at, resolved_at)

ESCROW_MILESTONES (id [PK], escrow_id [FK → ESCROWS.id], seq, title, amount, due_date, status, funded_txn_id [FK → TRANSACTIONS.id, UQ], closing_txn_id [FK → TRANSACTIONS.id, UQ]) (weak entity)

ESCROWS (id [PK], ref [UQ], buyer_id [FK → USERS.id], seller_id [FK → USERS.id], title, currency, status, created_at)

EXCHANGE_RATES (id [PK], base_currency, quote_currency, rate, as_of, source)

EXPENSE_GROUPS (id [PK], name, group_type, created_by [FK → USERS.id], created_at)

EXPENSE_SPLITS (id [PK], expense_id [FK → GROUP_EXPENSES.id], user_id [FK → USERS.id], share_amount, share_percent) (link relation for SPLIT_AMONG)

GROUP_EXPENSES (id [PK], group_id [FK → EXPENSE_GROUPS.id], paid_by [FK → USERS.id], description, category, amount, split_type, expense_date, created_at)

GROUP_MEMBERS (group_id [PK, FK → EXPENSE_GROUPS.id], user_id [PK, FK → USERS.id], joined_at) (link relation for MEMBER_OF)

GROUP_SETTLEMENTS (id [PK], group_id [FK → EXPENSE_GROUPS.id], from_user [FK → USERS.id], to_user [FK → USERS.id], amount, txn_id [FK → TRANSACTIONS.id, UQ], created_at)

IDEMPOTENCY_KEYS (id [PK], user_id [FK → USERS.id], idem_key, action, txn_id, created_at)

INVOICE_ITEMS (id [PK], invoice_id [FK → INVOICES.id], description, quantity, unit_price) (weak entity)

INVOICE_PAYMENTS (id [PK], invoice_id [FK → INVOICES.id], txn_id [FK → TRANSACTIONS.id, UQ], payer_id [FK → USERS.id], amount, created_at)

INVOICES (id [PK], number [UQ], issuer_id [FK → USERS.id], customer_id [FK → USERS.id], customer_name, title, currency, due_date, status, link_token [UQ], created_at)

IOU_REPAYMENTS (id [PK], iou_id [FK → IOUS.id], amount, txn_id [FK → TRANSACTIONS.id, UQ], created_at) (weak entity)

IOUS (id [PK], lender_id [FK → USERS.id], borrower_id [FK → USERS.id], amount, note, due_date, status, created_at)

LEDGER_ACCOUNTS (id [PK], code [UQ], kind, currency, name, balance)

LEDGER_ENTRIES (id [PK], txn_id [FK → TRANSACTIONS.id], phase, account_id [FK → LEDGER_ACCOUNTS.id], direction, amount, currency, created_at)

MERCHANTS (id [PK], user_id [FK → USERS.id, UQ], business_name, category, status)

MONEY_RULES (id [PK], user_id [FK → USERS.id], name, rule_type, target_user_id [FK → USERS.id], target_pocket_id [FK → POCKETS.id], amount, percent, round_to, threshold, frequency, next_run, status, fail_count, last_message, created_at)

NOTIFICATIONS (id [PK], user_id [FK → USERS.id], category, title, body, is_read, created_at)

OFFLINE_QUEUE (id [PK], merchant_user_id [FK → USERS.id], token_text, status, message, txn_id [FK → TRANSACTIONS.id], queued_at, synced_at)

OFFLINE_TOKENS (id [PK], token_id [UQ], user_id [FK → USERS.id], payee_user_id [FK → USERS.id], amount, expires_at, status, txn_id [FK → TRANSACTIONS.id, UQ], created_at)

PAYMENT_REQUESTS (id [PK], requester_id [FK → USERS.id], payer_id [FK → USERS.id], amount, currency, note, status, txn_id [FK → TRANSACTIONS.id], created_at)

POCKETS (id [PK], user_id [FK → USERS.id], name, pocket_kind, target_amount, target_date, account_id [FK → LEDGER_ACCOUNTS.id, UQ], created_at)

RATE_LIMITS (bucket [PK], hits, window_start)

RECURRING_ITEMS (id [PK], user_id [FK → USERS.id], kind, name, merchant, amount, currency, frequency, next_due, category, status, created_at)

REMINDER_LOG (id [PK], user_id [FK → USERS.id], ref_table, ref_id, due_date, sent_at)

RISK_ALERTS (id [PK], user_id [FK → USERS.id], txn_id, alert_type, severity, message, status, created_at)

ROLES (id [PK], name [UQ])

RULE_RUNS (id [PK], rule_id [FK → MONEY_RULES.id], run_key, status, message, txn_id, ran_at) (weak entity)

SETTINGS (skey [PK], svalue, description)

TIMELINE_EVENTS (id [PK], entity_type, entity_id, actor_id, event, note, created_at)

TOKEN_CHARGES (id [PK], token_id [FK → VIRTUAL_TOKENS.id], amount, status, reason, txn_id [FK → TRANSACTIONS.id], created_at) (weak entity)

TRANSACTIONS (id [PK], ref [UQ], type, status, from_wallet_id [FK → WALLETS.id], to_wallet_id [FK → WALLETS.id], amount, fee, currency, category, note, created_by [FK → USERS.id], refund_of [FK → TRANSACTIONS.id, UQ], ref_table, ref_id, created_at, completed_at)

TRANSFER_HOLDS (id [PK], txn_id [FK → TRANSACTIONS.id, UQ], release_at, status, reason, resolved_at) (weak entity)

USER_SETTINGS (user_id [PK, FK → USERS.id], language, hide_balance, notify_login, notify_transaction, notify_reminder, sim_biometric) (weak entity)

USERS (id [PK], name, email [UQ], phone [UQ], password_hash, pin_hash, role_id [FK → ROLES.id], status, failed_logins, locked_until, created_at)

VIRTUAL_TOKENS (id [PK], user_id [FK → USERS.id], wallet_id [FK → WALLETS.id], merchant_id [FK → MERCHANTS.id], token_hash [UQ], token_last4, per_charge_limit, monthly_limit, status, created_at)

WALLETS (id [PK], user_id [FK → USERS.id], currency, account_id [FK → LEDGER_ACCOUNTS.id, UQ], frozen, frozen_by, created_at)

Composite UNIQUE constraints (candidate keys beyond the primary key)

Relation	Unique columns	Why
APPROVAL_VOTES	request_id, approver_id	one vote per approver
BUDGETS	user_id, group_id, category	one budget per user, group and category
DEPENDENTS	guardian_id, dependent_id	one link per guardian and dependent
DEVICES	user_id, device_hash	one row per user device
ESCROW_MILESTONES	escrow_id, seq	ordered milestones per escrow
EXCHANGE_RATES	base_currency, quote_currency, as_of	one rate per pair and date
EXPENSE_SPLITS	expense_id, user_id	one share per member per expense
IDEMPOTENCY_KEYS	user_id, idem_key, action	duplicate-request protection
REMINDER_LOG	ref_table, ref_id, due_date	each reminder sent once
RULE_RUNS	rule_id, run_key	each rule run executes once
WALLETS	user_id, currency	one wallet per user and currency

CHECK constraints (8): positive amounts on TRANSACTIONS, LEDGER_ENTRIES, PAYMENT_REQUESTS, GROUP_EXPENSES, IOUS, RECURRING_ITEMS and ESCROW_MILESTONES, and a non-negative balance on

LEDGER_ACCOUNTS except the simulated funding and FX accounts.

7. From ERD to relational schema (mapping rules applied)

- **Strong entity** → one relation with the entity's attributes; the key attribute becomes the primary key (a surrogate integer id in every case).
- **Weak entity** → one relation with its own surrogate id and a foreign key (NOT NULL) to the owner entity; the identifying relationship is that foreign key.
- **One-to-many relationship** → the foreign key is placed on the N side (child). If the child must always have a parent the column is NOT NULL (total participation).
- **One-to-one relationship** → the foreign key is placed on the dependent side and made UNIQUE (for example WALLETS.account_id, TRANSFER_HOLDS.txn_id, MERCHANTS.user_id).
- **Many-to-many relationship** → a link relation containing both foreign keys plus the relationship attributes (MEMBER_OF, SPLIT_AMONG, APPROVER_OF, VOTES_ON). SPLIT_AMONG and VOTES_ON use a surrogate id plus a composite UNIQUE key of the two foreign keys.
- **Unary relationship** → REFUNDS: TRANSACTIONS.refund_of references TRANSACTIONS.id and is UNIQUE, so a transaction can be refunded at most once.
- **Two relationships between the same pair of entities** get separate foreign keys and role names, for example TRANSACTIONS DEBITS / CREDITS WALLETS (from_wallet_id, to_wallet_id) and LENDS / BORROWS between USERS and IOUS.
- **Polymorphic references** (TRANSACTIONS.ref_table/ref_id and TIMELINE_EVENTS.entity_type/entity_id) are not foreign keys; they are plain attributes used for navigation only, and are the one place where referential integrity is enforced by the application.

Consistency checks run against the live database

Check	Result
Tables in schema = entities + link relations	51 = 47 + 4: OK
Foreign keys drawn as relationships = foreign keys in schema	78 one-to-many/one-to-one + 8 link foreign keys = 86 (schema has 86): OK
Every entity appears in exactly one figure as a full entity	OK (47 entities across 11 figures)
Every relationship has a unique name	OK
Every relation has a primary key	OK